

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Uncovering insights from **3,900 transactions** to guide strategic business decisions.



Dataset at a Glance

3,900

Purchases

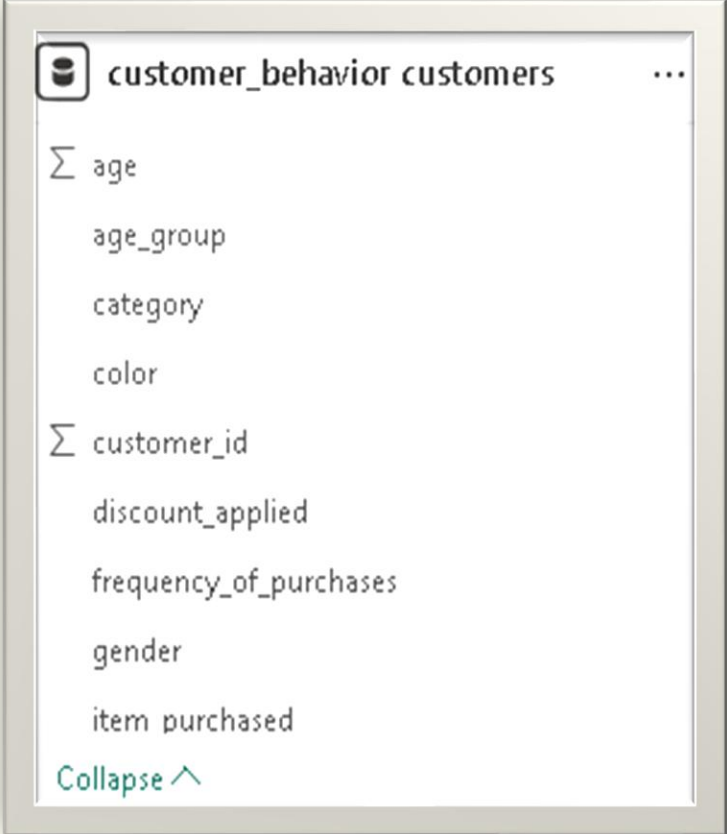
37

Missing Values

18

Columns

The dataset contains 3,900 customer shopping transaction records collected from a retail business. It includes customer demographics, product purchase details, payment methods, shipping preferences, subscription status, discounts, review ratings, and purchase frequency information.



A screenshot of a data management interface showing a table named 'customer_behavior' with 18 columns. The columns are listed in a scrollable area, with some grouped under a summation symbol (Σ). The columns are: age, age_group, category, color, customer_id, discount_applied, frequency_of_purchases, gender, and item purchased. A 'Collapse' button with an upward arrow is visible at the bottom of the column list.

customer_behavior	customers	...
Σ age		
age_group		
category		
color		
Σ customer_id		
discount_applied		
frequency_of_purchases		
gender		
item purchased		
Collapse ^		

Data Preparation in Python

Cleaned

Imputed missing Review Ratings
using category medians

```
import pandas as pd

df = pd.read_csv("C:/Users/PC/Desktop/Project/customer_shopping_behavior.csv")
df.head()

# Display basic information about the DataFrame
df.info()
df.describe(include='all')

# Check for missing values in each column
df.isnull().sum()

# Fill missing values in 'Review Rating' with the median rating for each category
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median))
```

Data Preparation in Python

Engineered Created `age_group` and `purchase_frequency_days`

```
# Create column purchase_frequency_days

frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 356,
    'Every 3 Months': 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping )
```

```
# Create a column age_group

label = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = label)
```

Data Preparation in Python

Loaded

Connected to MySQL for SQL analysis

```
import pandas as pd

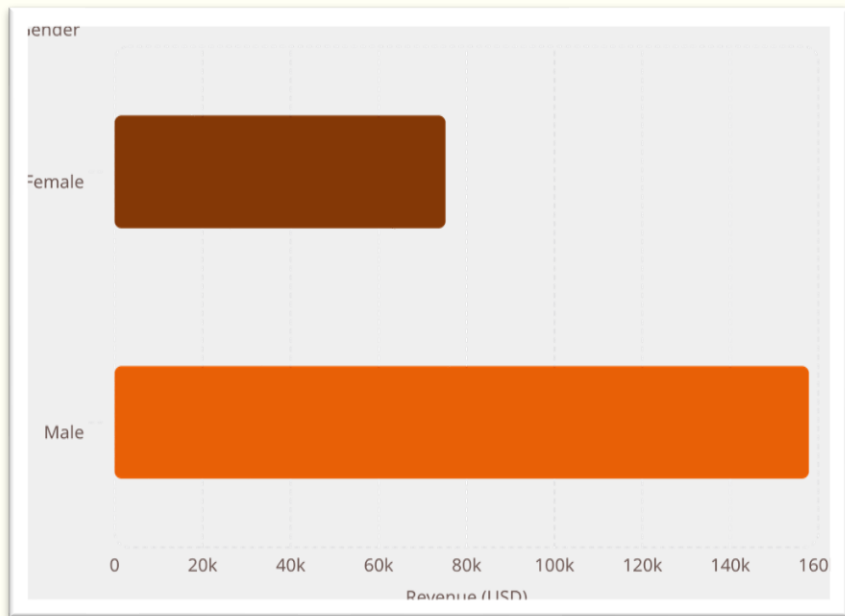
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Revenue by Gender



Key Finding

Male customers generated **2.1× more revenue** than female customers — a critical segment to understand and target.

Top Products & Discount Patterns

Top 5 by Rating

item_purchased	avg_review_rating
Sandals	3.82
Boots	3.79
Handbag	3.78
Hat	3.78
Gloves	3.78

Highest Discount Rates

item_purchased	discount_percentage
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Shipping & Subscription Insights

Express Shipping

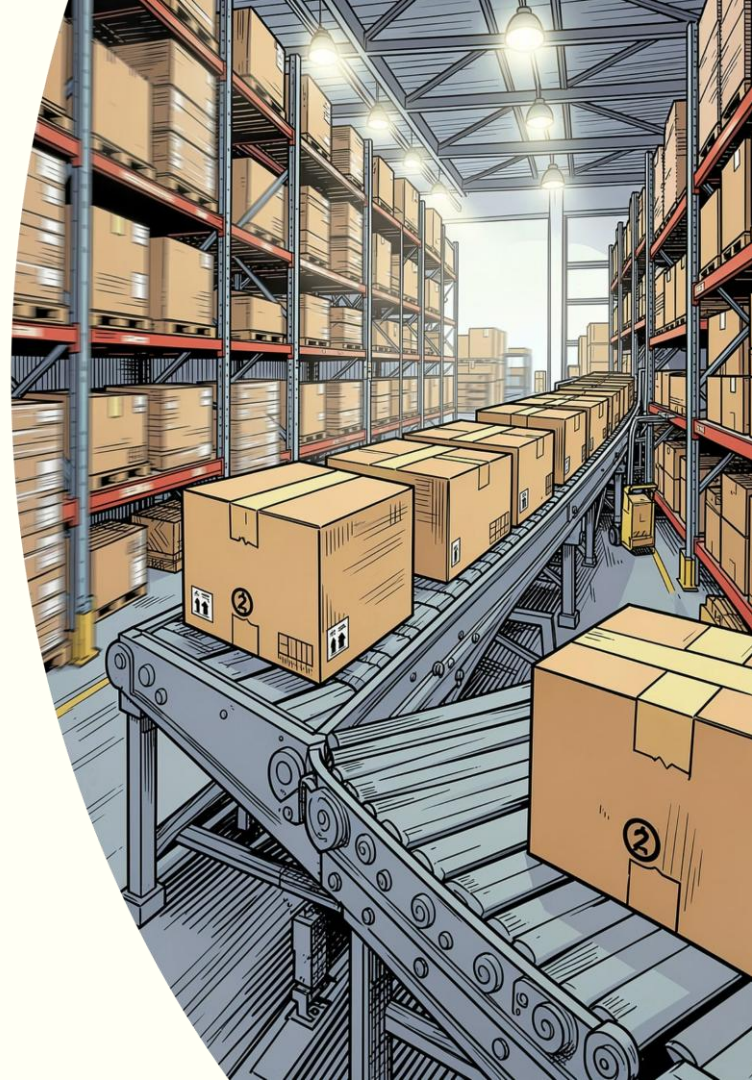
Avg spend **\$60.48** vs.
Standard \$58.46

Subscribers

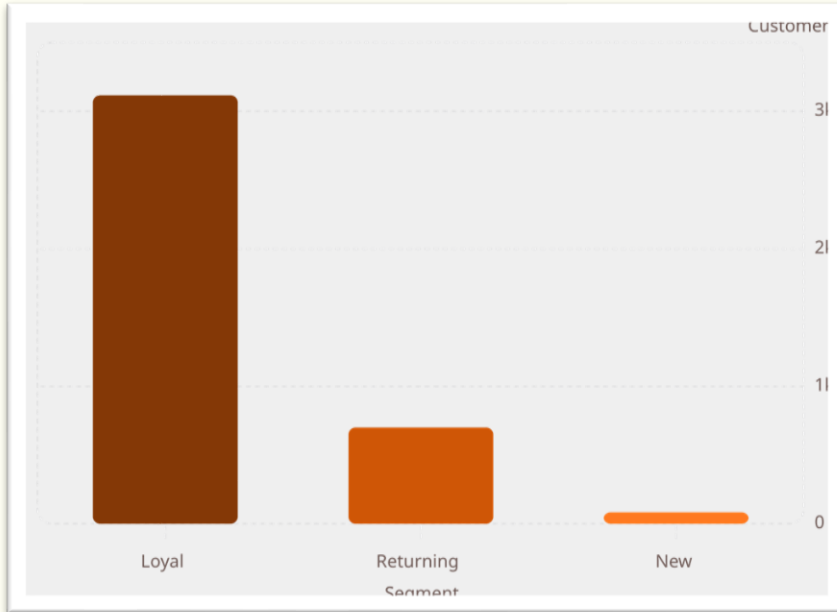
27% of customers · \$59.49
avg spend

Non-Subscribers

73% of customers · \$59.87 avg spend



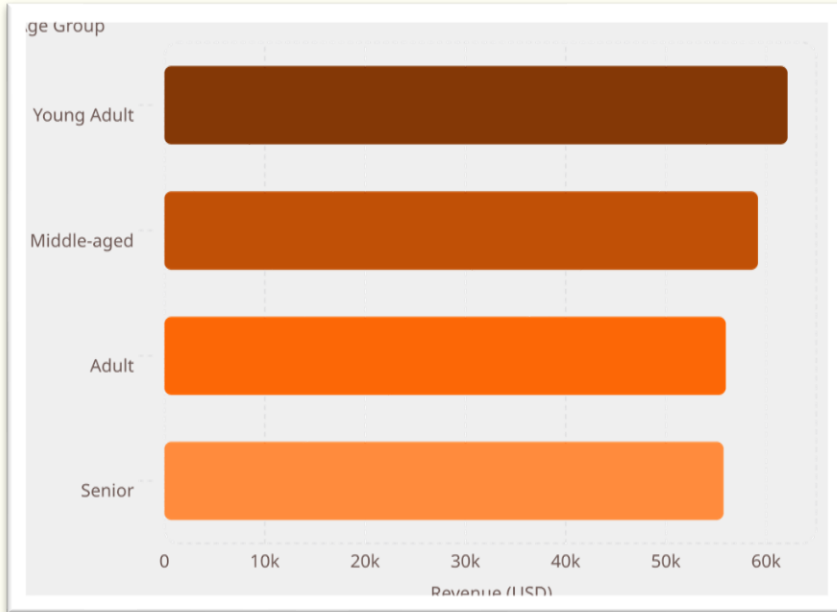
Customer Segmentation



Segment Breakdown

Over **80% of customers** are classified as Loyal — a strong retention base to leverage.

Revenue by Age Group



Key Insight

Young Adults lead revenue contribution. Revenue is relatively evenly distributed across all age groups.

Power BI Dashboard

Customer Behavior Dashboard

Subscription Status

No

Yes

Gender

Female

Male

Category

Accessories

Clothing

Footwear

Outerwear

Shipping Type

- ☐ 2-Day Shipping
- ☐ Express
- ☐ Free Shipping
- ☐ Next Day Air
- ☐ Standard
- ☐ Store Pickup

3.9K

Number of Customer

\$59.76

Average Purchase Amount

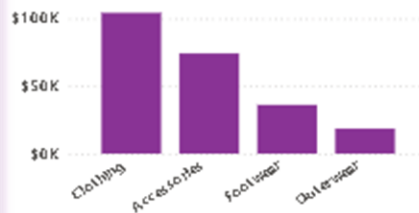
3.75

Average Review Rating

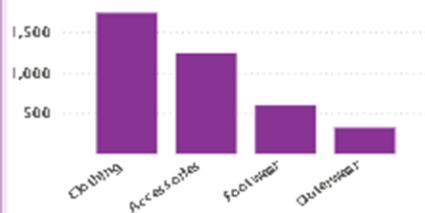
% of Customer By Subscription Status



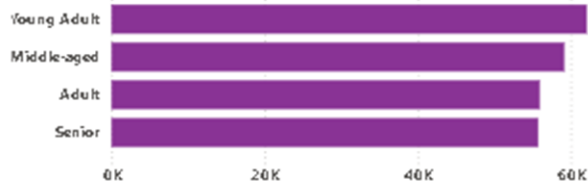
Revenue By Category



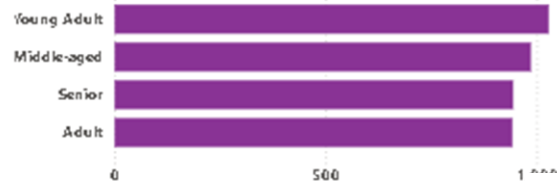
Sales By Category



Revenue By Age Group



Sales By Age Group





Business Recommendations

- Boost Subscriptions
Promote exclusive subscriber benefits
- Loyalty Programs
Reward repeat buyers toward "Loyal" segment
- Review Discount Policy
Balance sales boosts with margin control
- Targeted Marketing
Focus on high-revenue age groups & express-shipping users